Expert Forecasting with and without Uncertainty Quantification and Weighting: What Do the Data Say?

Roger M. Cooke[1], Deniz Marti[2], Thomas Mazzuchi[2]

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Abstract

Post-2006 expert judgment data has been extended to *530* experts assessing *580* calibration variables from their fields. New analysis shows that point predictions as medians of combined expert distributions outperform combined medians, and medians of performance weighted combinations outperform medians of equal weighted combinations. Relative to the equal weight combination of medians, using the medians of performance weighted combinations yields a *65%* improvement. Using the medians of equally weighted combinations yields a *46%* improvement. The *Random Expert Hypothesis* underlying all performance-blind combination schemes, namely that differences in expert performance reflect random stressors and not persistent properties of the experts, is tested by randomly scrambling expert panels. Generating distributions for a full set of performance metrics, the hypotheses that the original panels' performance measures are drawn from distributions produced by random scrambling are rejected at significance levels ranging from *E-6* to *E-12*. Random stressors cannot produce the variations in performance seen in the original panels. In- and out-of-sample validation results are updated.

Introduction:

Using expert uncertainty quantification (UQ) as scientific data with traceability and validation dates from (Cooke, 1987; Cooke et al., 1988; Cooke, 1991) under the name "Classical Model" or "Structured Expert Judgment". Distinguishing features are treating experts as statistical hypotheses and evaluating performance with respect to statistical accuracy and informativeness based on calibration variables (a.k.a. seed variables) from the experts' field to which true values are or become known post elicitation. The procedure results in combinations of experts' distributions (termed decision makers or *DM*s) using performance-based weighting (*PW*) derived from statistical accuracy and informativeness and also using equal weighting (*EW*). The reader is referred to Appendix 1 for specific details. This data enables the study of forecast accuracy, statistical accuracy, and informativeness of experts and of PW and EW combinations. Appendix 2 summarizes and updates in-sample validation results and Appendix 3 gives information and references for the post 2006 data.

Application highlights involved nuclear safety in the 1990s with the European Union and the United States Nuclear Regulatory Commission, fine particulates with Harvard University and the government of Kuwait in 2004-2005, food-borne diseases for the World Health Organization (WHO) in 2011-2013, ice sheet dynamics for Princeton University, Rutgers University and Resources for the Future (Bamber et al., 2019), and volcanic hazard levels in different parts of the world. The Classical Model was a key decision-support procedure during the prolonged eruption

[1] Resources for the Future, TU Delft, Dept Mathematics (retired); corresponding author
[2] George Washington University. Department of Engineering Management and Systems Engineering

on the island of Montserrat, West Indies in 1995-2018. Over the same period, expert elicitations using the Classical Model have informed many issues of public health policy, civil aviation safety, fire impacts on engineered structures, and earthquake resistance of critical utility facilities.

Validation is the cornerstone of science. A special issue on expert judgment (Cooke & Goossens, 2008) focused on this issue. Colson and Cooke (2017) gave an extensive review of validation research and applied the cross validation code of Eggstaff et al. (2014) to the *33* professional studies post-2006 available at that time. These studies comprised *320* experts assessing *369* calibration variables. These numbers have since grown to *49* studies involving *530* experts and *580* calibration variables. Recently Wilson (2017) and Wilson and Farrow (2018) used the pre-2006 data to study expert dependence and its effect on validation both with regard to uncertainty quantification and point predictions. They documented inferior performance of equal weighting but note that very few experts exhibit large error correlations between questions. Their concern that expert performance may not be a persistent property of experts is addressed here with the updated dataset. Their researches into dependence can hopefully be extended with the present data.

The best current summary of the Classical Model, applications and validation research are published by Colson and Cooke (2017 and 2018; see esp. online supplements[3]).The reader is referred to these sources for older publications. Post-2006 studies are better resourced, better documented and more uniform in design than the older studies. They provide a unique data base for comparing expert predictions, and predictions of combinations of experts, with true values from the experts' fields. All data, both pre and post-2006 are available at http://rogermcooke.net/.

The Classical Model aims at uncertainty quantification and the underlying performance measures are designed to reward good uncertainty assessors. It is sometimes said that uncertainty quantification is not indicated when the interest is only in point predictions. This unique expert judgment data resource enables that idea to be rigorously tested. In section 2 it is shown that uncertainty quantification (UQ) improves point predictions in the following sense: combining expert median assessments is inferior to taking the median of a combination of expert UQ's. Equally weighted combination of experts' "best guesses" is the default predictor when UQ is not employed. Against this benchmark (treating experts' medians as "best guesses"), using the medians of performance weighted combinations yields a *65%* reduction in prediction error . Using the medians of equally weighted combinations yields a *46%* reduction.

Section 3 briefly reviews and updates cross validation results. Section 4 addresses the concern of Wilson (2017), Wilson and Farrow (2018). We test the hypothesis that fluctuations in expert performance can be explained by random stressors during the elicitation, termed the Random Expert Hypothesis (REH). If REH were true then panel performance should be statistically unaffected by randomly re-allocating the individual assessments over the panel members. Note that performance-blind combination methods such as equally weighting expert uncertainties or combining expert quantiles with equal weight are invariant under randomly scrambling the expert assessments. Performance weighted combinations on the other hand depend on identifying the best performing experts in a panel and assigning them high weight. The hypotheses that panel metrics

[3] See https://www.sciencedirect.com/science/article/pii/S0951832017302090?via%3Dihub#s0065

such as *statistical accuracy of best expert in a panel*, and *standard deviation of experts' performance* are statistically unaffected by randomly scrambling expert assessments are rejected at significance levels ranging from *E-6* to *E-12*. These tests are more powerful than the previous out-of-sample cross validation tests.

The driver behind virtually all these applications is the validational aspect of the Classical Model. Although the psychological community has long drawn attention to cognitive biases inherent in expert UQ, and despite a robust interest in validation research, there are barriers to the use of performance measures. The conclusion speculates on possible explanations for this.

2. Point prediction; a co-benefit of expert UQ

Statistical accuracy and informativeness are performance metrics for quantifying uncertainty. There is nothing in these metrics that rewards proximity of the medians to the true values. If these performance metrics enable more accurate predictions of the true values, then this is a collateral benefit, or co-benefit of performance weighting. The Median Distance for variable i from a distribution with $Median_i$ is defined as $MD_i = |\,(Median_i - true\ value_i)\,|$, where $true\ value_i$ is the true value of the calibration variable i.

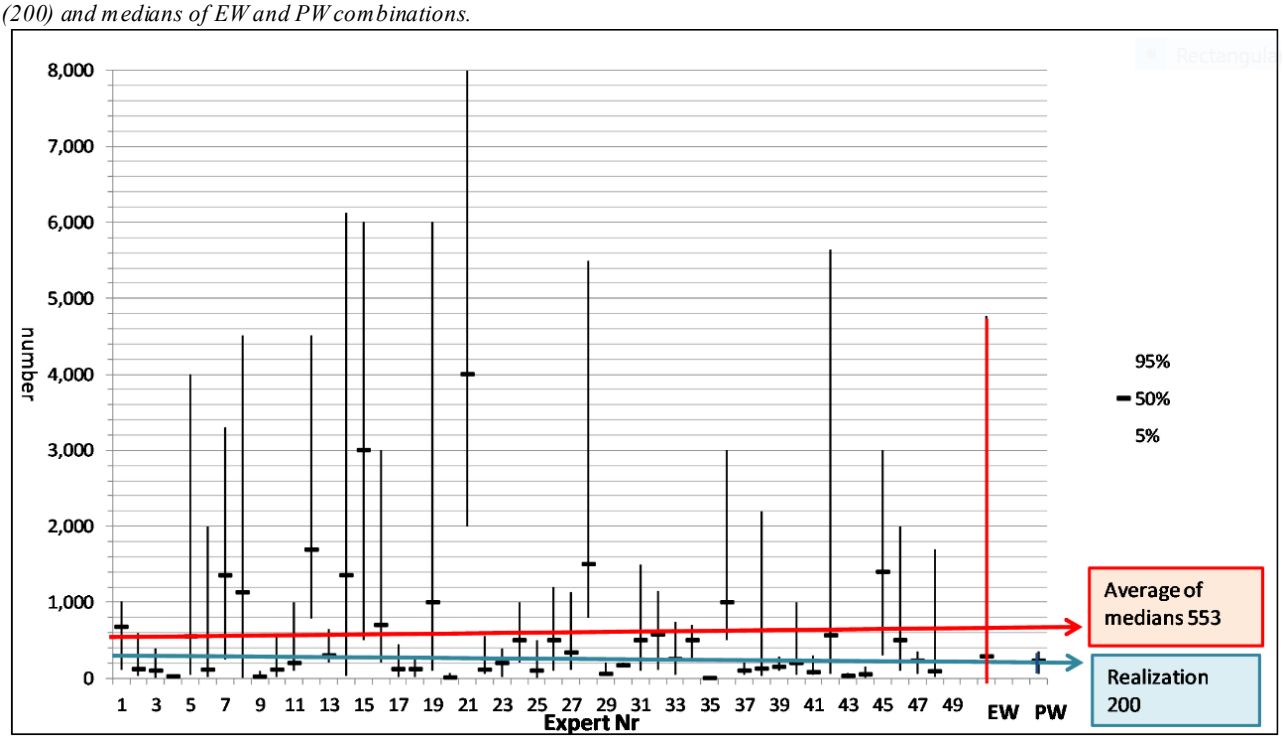
MD_i is dependent on the scale of variable i; changing from, say, meters to kilometers will affect the value of MD_i. To aggregate over variables with different scales, the scale dependence must be removed. To compare the proximity of the medians of *PW* and *EW* to the realizations, taking the ratio of *MD* for *EW* and *PW* (denoted as *EWMD* and *PWMD*, respectively) per variable removes the scale dependence. These ratios are then aggregated over all variables in a study by taking the geometric mean (geomean):

$\frac{EWMD}{PWMD} = \left[\prod_{i=1}^{N} \frac{EWMD_i}{PWMD_i}\right]^{1/N}$ where N is the number of calibration variables.

The geomean is appropriate for aggregating ratios since the geomean of inverse ratios is the inverse of the ratios' geomean and the geomean of ratios is the ratio of geomeans. However, for *10* of the *580* predictions, the item specific PW (PW_i, see Appendix 1) median was actually equal to the realization, making the above ratio infinite (this does not happen for other predictors). These *10* variables were therefore excluded from the comparisons, although this is prejudicial to PW_i. For a few other variables $|PW_i - true\ value_i|$ is so small as to render the above ratio very large. While these variables would dominate the average of this ratio, the geomean is relatively insensitive to large outliers.

While the mean of a linear combination of distributions is the linear combination of their means, the same does not hold for the median. A recent elicitation on food borne illness pathways at the CDC illustrates this difference. One of the *14* calibration questions was: "*Between Jan. 1, 2016 and Dec. 31, 2016, a total of 11,277 samples of raw ground beef from 1,193 establishments were tested for Salmonella spp. Of these samples, how many tested positive for Salmonella spp?* ". The *5, 50* and *95* percentiles of the *48* participating experts, EW and PW combinations are shown in Figure 1, as is the true value *200*. The medians of the EW and PW combinations are quite close to *200* whereas the average of medians is *533*.

Figure 1: Responses of 48 experts in a recent CDC elicitation, showing the average of medians (533), the true value or Realization (200) and medians of EW and PW combinations.



Although quantile averaging is usually done unawares, Lichtendahl et al. (2013) have recommend this as opposed to "averaging probabilities". This has been shown to produce highly overconfident results (Colson & Cooke, 2017). A brief explanation suffices: Consider two experts with 5th and 95th percentiles of *[0,1]* and *[10, 11]* respectively. Averaging their percentiles yields a *90%* confidence interval of *[5, 6]*; equally narrow but disjoint from each expert's confidence interval. Experience in expert judgment shows that such situations are not uncommon. "Averaging the probabilities" requires knowledge of the entire distributions. It is more complex but produces distributions more evenly spread over the interval *[0, 11]*. For purposes of combining distributions, averaging quantiles is a very common and rather severe mistake.

Finding simple point predictions does not require combining distributions. One could simply take a simple linear combination of the experts' medians rather than first combining their distributions. People will continue combining quantiles in any case; therefore it is useful to assess the ensuing loss of performance. *PWMDQ* and *EWMDQ* denote respectively the performance weighted and equally weighted combinations of the experts' medians ("Q" denotes quantile averaging). *PWiMD* represents the "high end" predictor based on item specific performance weights (see Appendix 1). *EWMDQ* is the "low end predictor". It is the predictor most often applied for expert judgment and is therefore chosen as benchmark. Eliciting only a "best guess" should be discouraged. This invites confusion as to whether the mode, median or mean is intended and there is no reason to think that performance of "best guesses" in point prediction is any better to *EWMDQ*.

Figure 2 plots *EWMDQ/EWMD* and *EWMDQ/PWiMD* per study. Values greater than *1* indicate superior predictions relative to *EWMDQ*. The geomean over all 570 predictions of the prediction

error ratio *EWMDQ / PWiMD* is *1.65* and that of *EWMDQ / EWMD* is *1.46*. On aggregate, predictions of *PWiMD* are *65%* closer to the truth than those of *EWMDQ*, for *EWMD* the improvement is *46%*. Even if one is only interested in expert based point predictions, it is preferable for the experts to quantify their uncertainty and to combine their distributions with equal weights. It is better still to measure performance as probability assessors and form weighted combinations of their distributions. These studies have no common experts and may be considered independent. However, taking the geomean of forecast errors per study, then taking the geomean of these geomeans over all studies over-weights the studies with smaller numbers of calibration variables and would result in slightly lower improvements (1.60 and 1.41 respectively).

To compare individual forecasts with realizations the absolute forecast error must be made scale independent. Table 1 gives the absolute dimensionless forecast errors *|(forecast – true value)/true value|* for each of the 569 forecasts for which the realization is non zero. A perfect score would be zero. Averages and standard deviations are strongly affected by large outliers. The geometric average (geomean) is insensitive to large outliers. Depending on how one values large outliers, the laurels go to medians of item specific performance weighting *(PWi)* or medians of non- optimized global performance weighting (*PWn)*. In any event, the benchmark of equal weighted combinations of experts' medians is the poorest performer. If one eschews UQ one is condemned to quantile averaging with a marked penalty is forecast error. UQ without performance weighting avoids the effort of finding suitable calibration variables, but suffers a performance penalty.

Figure 2: Ratios of EWMDQ / EWMD (red) and EWMDQ / PWiMD (blue) per study

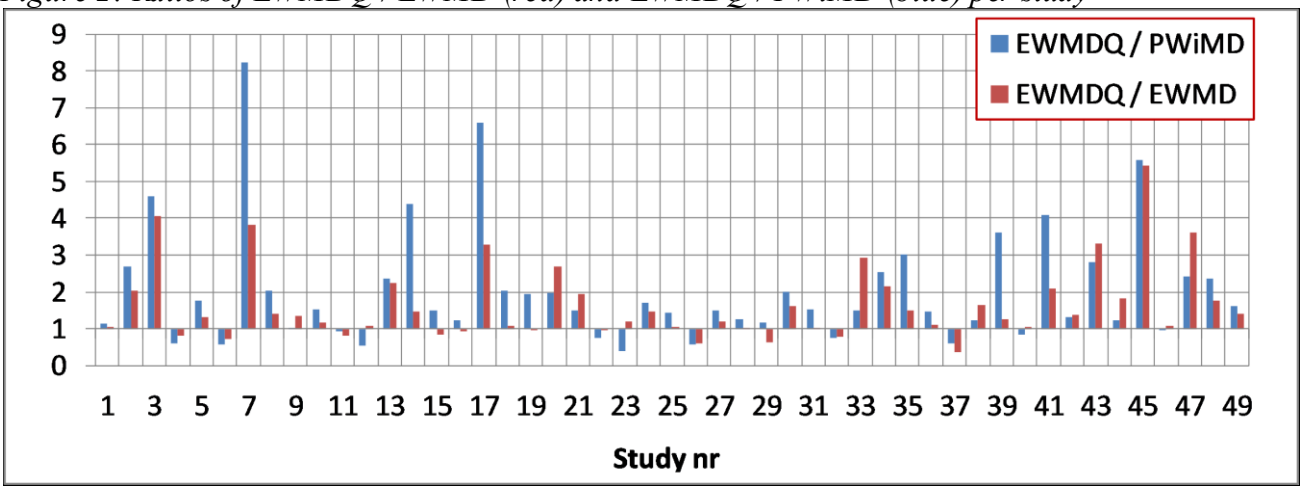
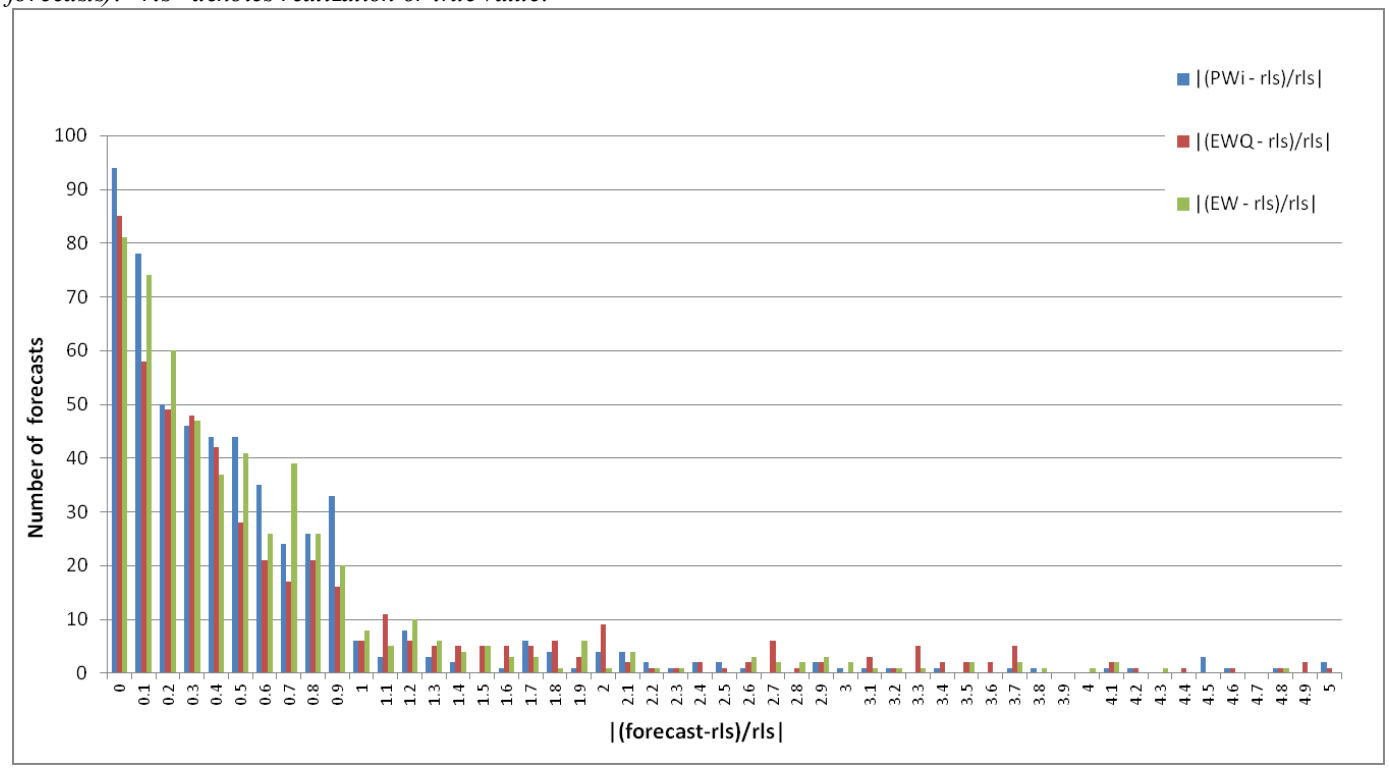


Table 1: *Average and standard deviation of absolute dimensionless forecast errors for item specific performance weights (PWi) global performance weights (PWg), non-optimized global performance weights (PWn), equal weights (EW), Performance weighted average of medians (PWQ) and equal weighed average of medians (EWQ) (for definitions see Appendix 1). "rls" denotes "true value" or realization.*

	|(PWi - rls)/rls|	|PWg - rls|/rls	|PWn - rls|/rls	|(EW - rls)/rls|	|PWQ - rls|/rls	|(EWQ - rls)/rls|
Ave	2.2	2.7	2.3	3.8	278.6	1472.3
Stdev	11.8	16.0	14.7	45.2	5646.8	33299.8
Geomean	0.38	0.40	0.37	0.43	0.42	0.63

The mass functions for forecasts whose dimensionless absolute error is less than *5* are shown in Figure 3. For PW_i *94* of the *569* forecasts has absolute dimensionless forecast error less than 0.1, for EW this number is *81*.

Figure 3: Dimensionless absolute forecast errors less than 5, PWi,(541forecasts) EW(534 forecasts) and EWQ(498 forecasts). "rls" denotes realization or true value.



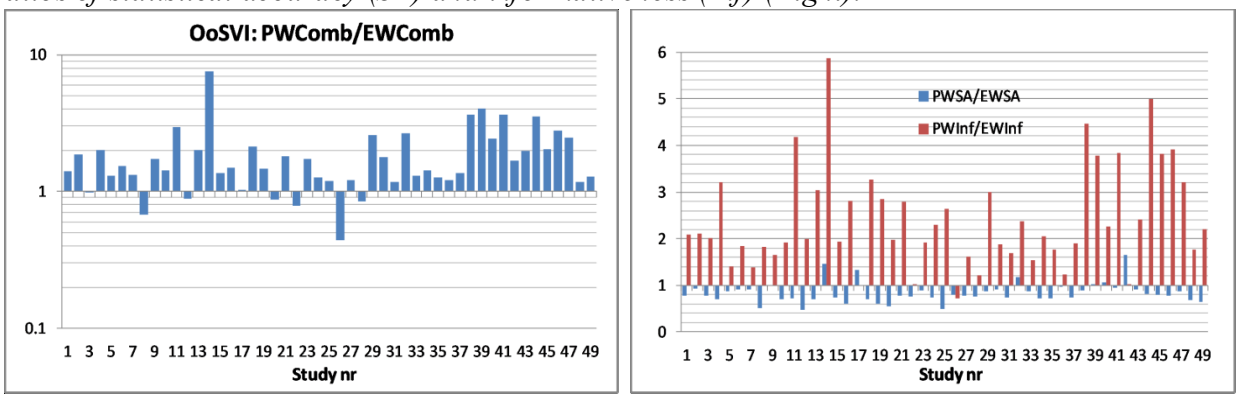
3. Out-of-sample Cross Validation

Considerable effort has been devoted to validating expert judgments. Unless the variables of interest can be observed shortly after completion of a study, out-of-sample validation comes down to cross validation. The calibration variables are split into a training set for initializing the *PW* and a test set for comparing *PW* and *EW*. The sets on which the performance weights are derived and evaluated are thus disjoint. Early efforts with different methods produced uneven results as documented by Colson and Cooke (2017). Col Eggstaff and colleagues produced out-of-sample validation code and applied this to all pre 2008 expert data. It is the only code to our knowledge that has been benchmarked against the standard expert judgment software. This code was applied to 33 post 2006 studies by Colson and Cooke (2017) and is here extended to the current set of 49 post 2006 studies. Understanding the strengths and limitations of cross validation is essential to appreciate the contributions of testing the REH.

Many issues involved in choosing the training and test set sizes are discussed in (Colson and Cooke 2017), to which we refer the interested reader. The upshot is that using *80%* of the calibration

variables as a training set best balances the competing goals of resolving expert performance on the training set and resolving the performance of combinations on the test set. The training set then has enough statistical power to reduce the variance in the expert weights thereby rendering the performance weights similar to the weights based on all calibration variables. The test set loses statistical power for resolving the *PW* and *EW DM*s, but with *10* calibration variables statistical accuracy scores for assessments of 5th , 50th , and 95th percentiles still vary by a factor *31*. Moreover, higher resolution is of no value if the *PW DM* is very volatile and unlike the *PW DM* of the full study. Of course the actual sizes of the training and test sets vary with the total number of calibration variables. The *80%* split makes it easier to pool the results from all studies. With *10* calibration variables there are *45* distinct *8*-tuples of calibration variables to be used as training sets. Performance is scored on the *2* remaining variables. The statistical accuracy, the informativeness and the combined score (the product of the former two) are averaged over the *45* different test sets. Colson and Cooke (2017) show that the average ratio of combined scores for *PW* and *EW* is indistinguishable from the ratio of average combined scores for fixed training set size. The ratio of combined scores based on 80% of the calibration variables is called the *"Out of sample Validity Index (OoSVI)."*

Figure 4: Ratios of combined scores PWcomb / EWcomb averaged over all training sets sized at 80% of the calibration variables for 49 post-2006 studies (left). Combined scores factored into ratios of statistical accuracy (SA) and informativeness (Inf) (Right).

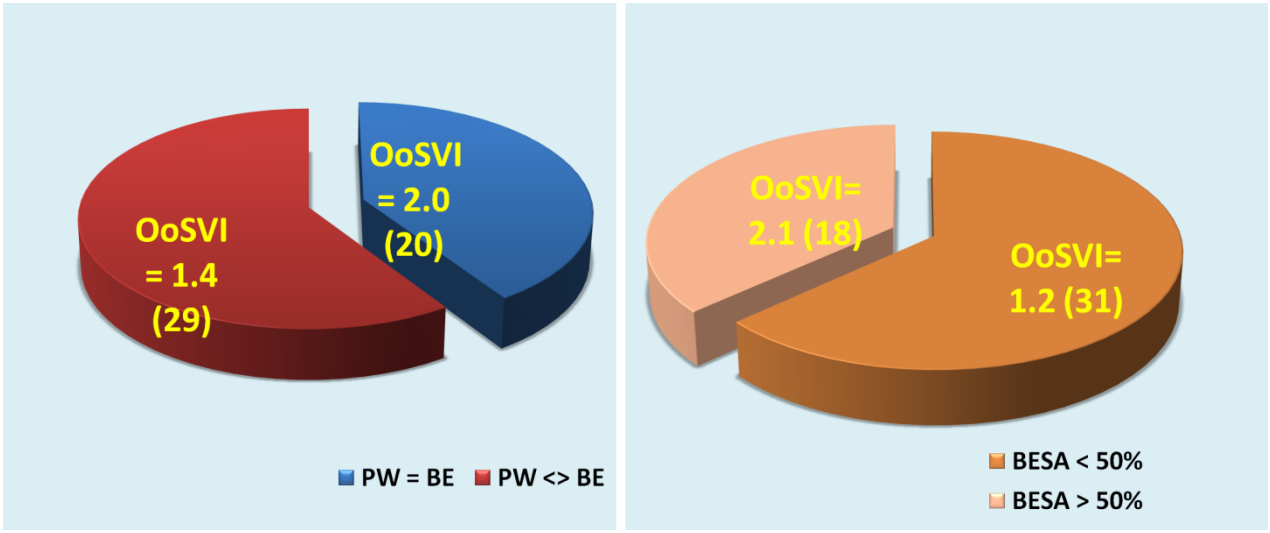


For *42* of the *49* studies the ratio *PW*comb / *EW*comb is greater than *1*. Under the null hypothesis that there is no difference between *PW* and *EW*, the probability of seeing *42* or more ratios greater than *1* is *1.8E–7*. The right panel of Figure 4 shows that *PW* suffers a modest out-of-sample penalty in statistical accuracy, which is more than compensated by a boost of informativeness. Lest this boost seem "small" it is well to note that a factor 2 increase in informativeness corresponds roughly to halving the length of the 90% confidence bands. If an assessor is perfectly statistically accurate, his/her statistical accuracy score is uniformly distributed on the interval [0, 1] with expected value 0.5. The mean out-of-sample statistical accuracy score (p value) for EW is 0.54, while that of PW is 0.43. In a nutshell, CM is able to boost informativeness substantially without sacrificing statistical accuracy.

As in (Colson and Cooke 2017), the features which best explained the differences in the *OoSVI* were studied. The results echo those earlier findings; if *PW* concentrates all weight in the best expert *(BE)* the overall geomean of *OoSVI* for all studies (*1.63)* splits into *2.0 (PW=BE)* and *1.4*

(PW ≠ BE). Similar results are obtained by splitting into studies in which BE's statistical accuracy is above 0.5*(2.1)* or below 0.5 *(1.2)*. Other features such as number experts, number of calibration variables and plenary versus individual elicitation had less effect. The quality of the best expert is the main determinant for *OoSVI*. The rank correlation between *OoSVI* and the in-sample ratio *(PW/EW)* of combined scores is *0.5*; these measures are related but not identical for reasons addressed in the following paragraph.

Figure 5: Breakdown of OoSVI scores in Figure 4 according to whether PW was identical with the best expert (PW=BE; left) and whether BE's statistical accuracy was greater than 50% (BESA>50%; right). The Geomeans of the 49 studies are shown for the of ratios of combined scores PWcomb / EWcomb evaluated study-wise for training sets sized at 80% of the calibration variables



Note. OoSVI refers to out-of-sample-validity index. BE denotes the best expert, BESA denotes the statistical accuracy of the best expert.

Cross validation is essential for demonstrating the value of *PW* relative to *EW* for out-of-sample prediction. The necessity of splitting the calibration set into training and test sets exacts a toll that is illustrated with the "Ice Sheet 2018" study ('ICE_2018;' see Appendix 3) involving *20* experts and *16* calibration variables. With a training set of *13* (*80%* of *16),* there are *560* distinct training sets. *8* of the *20* experts were weighted on at least one of these sets. For *7* of these *8*, the difference between their maximal and minimal weight was *1*; that is, their weights vacillated between *0* and *1*. The *PW* combinations evaluated on the 3 test variables still exhibit volatility and deviate from the *PW* of the entire study. The most we can say is that the *OoSVI* compares the score of *EW* with the scores of a swarm of *PWs*, which loosely resemble the *PW* of the full study. The cross validation data validates the performance weighting method, but not a specific *PW* combination.

4. The Random Expert Hypothesis

Expert judgment studies differ from simple surveys in that experts undergo a credentialing process before being empanelled. It is natural to believe that, once empanelled, experts should have an equal share in determining the panel output. This in turn leads naturally to aggregation schemes

based on equal weighting. Against this viewpoint is a mounting body of evidence that, as regards uncertainty quantification, experts' performance is very uneven. About three quarters of the *530* experts reviewed in Appendix 2, considered as statistical hypotheses, would be rejected at the 5% level. From this perspective, avowals of degrees of confidence in outcomes based on experts' judgments are susceptible to and in need of validation. This is hardly new. The US Defense Intelligence Agency adopted expert subjective probabilities in 1976 and dropped them a few years later for lack of validation[4] . Nonetheless the theoretical debate over whether and how to validate expert subjective probabilities continues.

Recently a new approach to validation has emerged (Marti et al., 2019). Whereas cross validation is hampered by the two-sided loss of statistical power caused by splitting calibration variables into training and test sets, the new approach does not focus on the performance of a combination of experts. Instead, it focuses on the expert performances themselves and investigates the assumption underlying all performance-blind approaches; namely that performance measures are unable to make meaningful distinctions in experts' performances (the Random Expert Hypothesis *REH)*. This may be because the experts are all equally good or equally bad. It may also be that any putative differences are swamped by the noise inherent in expert judgment: experts are influenced by random stressors, they may get distracted or fatigued, their performance is affected by the particular choice of calibration variables, etc. Concerns in this regard have been raised by Winkler et al. (2018) and Wilson (2017).

Note that if experts' differences in performance are caused by factors local to the elicitation of calibration variables, then the cross validation results of the previous section might not extend to the variables of interest. Further, the low power of the test set means that the *DM*s' statistical accuracy scores are poorly resolved and thus extra susceptible to random stressors. The new approach to validation focuses directly on *REH* without the intermediary of performance based combinations. Indeed, the *REH* itself carries a heavy proof burden and can be tested using expert performance provided by our *49* post-2006 studies. Intuitively, if performance differences are the result of noise, then randomly re-allocating the experts' assessments among the panel members will randomly re-distribute the random stressors. The fluctuations in performance produced in this way should envelope the performance in the original panels. It emerges that tests of *REH* are much more powerful than the cross validation tests.

To make this idea precise, consider a *random scramble* of an expert panel composed of *15* experts and *10* calibration variables. We create '*scrambled expert 1'* by randomly choosing an assessment without replacement from one of the *15* experts for the first variable, a second random draw without replacement gives the second assessment for '*scrambled expert 1*' and so on. '*Scrambled expert 2*' chooses his assessments in a similar way from the assessments not chosen by '*scrambled expert 1*'. The final scrambled expert, '*scrambled expert 15*' gets the leftovers. In this scrambled panel, we can measure the statistical accuracy (*SA*) and informativeness (*Inf*) of each expert, we can measure the combined scores ($SA \times Inf$), we can compute the average scores, the maximum and minimum and the standard deviation of the scores.

[4] See https://www.resourcesmag.org/common-resources/iceman-cometh/ .

For each study, we repeat the scrambling *1000* times and build up a distribution for each performance metric. This distribution reflects the variation we should see in that performance metric if experts' performance differences were due only to random stressors. Suppose we compute the average *SA* for experts in each of the *1000* scrambled panels for a given study. The *REH* now asserts that the average *SA* in the original panel could just as well be any of 1000 averages in the scrambled panels. There should be a *50%* chance that the original average *SA* is above the median of the *REH* distribution, a *5%* chance that it is above the 95th percentile of the *REH*, etc. Thus, *REH* expects that in *2.45* of the *49* studies, the original average *SA* should fall above the 95th percentile of the *REH* distribution. Actually, this happens in *20* of the *49* studies. The probability of *20* or more studies falling above the 95th percentile if *REH* were true is *6.6E–14*. *REH* fails if the differences in the experts themselves in the original panel are greater than what can be produced by scrambling the experts.

Note that random scrambling will have no effect on the *EW* combination. Assuming that *EW* is at least as good as *PW* implies *REH*. In consequence (modus tollens), if *REH* is (statistically) rejected, then so is the assumption that *EW* is at least as good as *PW*. In this sense, *REH* provides a more powerful test of the assumption underlying the use of *EW*. The same holds for the "averaging quantile" approaches (Lichtendahl et al., 2013) or indeed any approach which is performance-blind. If all experts in a panel are "equally good" or "equally bad," then *REH* may actually be true for that panel. The use of *PW* depends on the fact that such panels are in the minority. Testing *REH* on a set of cases allows us to gauge the size of that minority.

The data has been standardized in ways that do not affect the *REH*; experts who did not assess all calibration variables were dropped, reducing the number of experts from 530 in Figure A2.1 to 526. All background measures are converted to uniform (most calibration variables already have uniform backgrounds). Whereas some studies assessed the 25th and 75th percentiles in addition to the 5th , 50th and 95th percentiles, Marti et al. (2019) showed this had no effect on the REH so only the 5th , 50th and 95th percentiles are used.

For each of the *49* studies, the following eight performance metrics shown in Figure 6 are computed for the original panel and for each of the *1000* scrambled panels:

1. Panel Average Statistical Accuracy
2. Panel Max Statistical Accuracy
3. Panel Standard Deviation of Statistical Accuracy
4. Panel Min Statistical Accuracy
5. Panel Average Combined Score
6. Panel Max Combined Score
7. Panel Standard Deviation of Combined Score
8. Panel Min Combined Score

For metrics not involving the Min, we are interested in the quantile of the *REH* distribution realized by the metric in the original panel. For metrics 4 and 8 we are interested in the complimentary quantile, that is, the fraction of the *1000* scrambled panels in which the original minimum is *lower* than the scrambled minima. This is done so that all metrics have the same sense; numbers close to

1 are favorable for *PW*. We test the *REH* against the alternative that experts' performance differences are too large to be explained by random stressors and high values are critical.

If *REH* were true, that is, if the original panel's metrics were really drawn from the *REH* distributions, then the quantiles in Figure 6 should be uniformly distributed on the interval *[0, 1]*. The number of bars above the value *0.5* should be statistically equal to the number below *0.5*. The "amount of color" above *0.5* should statistically equal the amount below *0.5*.

There are two simple tests for the *REH* hypothesis. The binomial test simply counts the number of values greater than *0.5* for each metric and reports the *p* value for the corresponding null hypothesis: *the probability that half of the random panels outperform the original panel metric is 0.5*. The binomial test does not consider how far above or below *0.5* the metrics are. The *sum test* simply adds the *49* original panel quantiles for each metric. Under *REH* this sum should be (very, very nearly) normally distributed with mean *49/2 = 24.5* and standard deviation $(49/12)^{1/2} = 2.02$. For example, 'Average Statistical Accuracy' in the original panel exceeds the median of the *REH* distribution for 'Average Statistical Accuracy' on *42* of the *49* studies. If the probability of exceeding the median were really *0.5,* the probability of seeing *42* or more "successes" would be *1.81 E-7.* Summing the original panels' *49* realized quantiles in the *REH* distribution for 'Average Statistical Accuracy' yields *38.36.* The probability that a normal variable with mean *24.5* and standard deviation *2.02* exceeds *38.36* is *3.48 E-12*. The sum test is much more powerful than the binomial test. Table 2 collects the results for the binomial and sum tests. Interestingly, departures from *REH* seem more severe for the panel minima than for the panel maxima. In other words, REH has more difficulty reproducing the very low scores than the very high scores. Of course both departures are highly significant. Suppose we reject *REH* for each of the *49* studies. The sum of the *p* values (1 – percentile of original panel) gives the expected number of false rejections. This number is *49 – 38.36 = 10.64*. We might expect that REH is true in one fifth of the studies.

Figure 6: Quantiles of the REH distributions realized by the performance metrics for Statistical Accuracy (top) and Combined Score (bottom) by the original expert panels, for 49 studies.

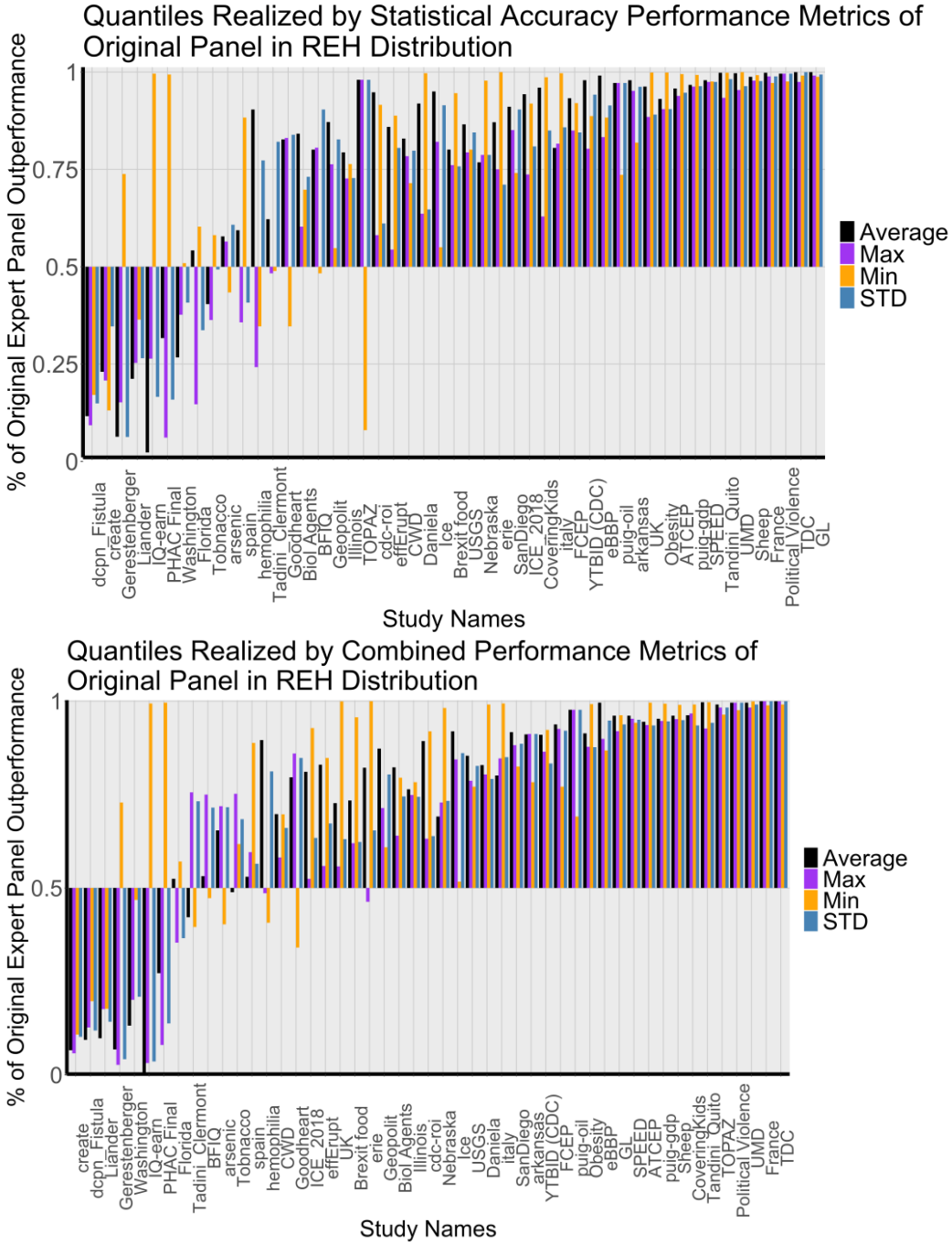


Table 2: p values at which REH is rejected for the seven performance metrics

p* values for tests of *REH		
	Binomial test	**Sum test**
Average Statistical Accuracy	1.81E-07	3.49E-12
Standard Deviation of Statistical Accuracy	4.63E-06	6.76E-10
Maximum Statistical Accuracy	2.35E-04	2.85E-06

Minimum Statistical Accuracy	4.63E-06	2.99E-11
Average Combined score	9.82E-07	3.21E-09
Standard Deviation of Combined Score	9.82E-07	3.96E-08
Maximum Combined Score	1.92E-05	5.85E-07
Minimum Combined Score	4.63E-06	6.67E-12

Whichever test we use, the notion that putative differences in expert performance are due to random stressors is overwhelmingly rejected. Table 3 examines the influence of the number of experts and number of calibration variables on the performance metrics.

Table 3: Spearman's rank correlation between number of experts and number of calibration variables and percentile scores.

	Quantile of Avg *SA*	Quantile of STD *SA*	Quantile of Max *SA*	Quantile of Avg. Comb	Quantile of STD Comb	Quantile of Max Comb
Rank Correlation to # Experts	0.24	0.13	0.03	0.18	0.05	-0.04
Rank Correlation to # Variables	0.40	0.39	0.39	0.31	0.34	0.31

Note. Avg denotes average, *SA* is for Statistical Accuracy, STD is the standard deviation, Comb. is for combined score, Max is for maximum.

With *49* samples, a rank correlation of 0.24 is significant at the 5% level. As seen in Table 3, the number of experts is not strongly associated with any of the metrics. The number of calibration variables does appear to exert some influence. Table 3 implies that more calibration variables tend to make the differences between the performance of original experts and randomly scrambled experts greater.

Conclusions

Experts exhibit undeniable differences in their ability to quantify uncertainty. These differences can readily be measured and used to improve uncertainty quantification and also point predictions. The evidence is overwhelming. However, there are two significant hurdles to applying performance based combinations: (1) time and effort required for performance measurement and (2) numeracy demands on the analyst.

In most applications, the greatest time and effort is spent in formulating clear questions with operational meaning which address the issues at hand. It is useful to think of expert judgment as a way of obtaining (probabilistic) results from experiments or measurements which are feasible in principle but not in practice. Describing a "thought experiment" is the best way of making absolutely clear what one is asking. These efforts should be made in any case, but the need for operational meaning is easier to ignore if there are no calibration questions. Having formulated questions with clear operational meaning facilitates finding calibration variables from the experts' field. The impractical experiments or measurements often suggest experiments and or measurements which are already performed though not published. Often, as in the recent 'Ice Sheet 2018' study, more time is spent agreeing on the best set of calibration variables than in generating them. That said, finding good calibration variables does require a deep dive in the subject matter and is greatly aided by having a domain expert on the analysis team. Quigley et al (2018) provide guidance on finding calibration variables.

One-on-one interviews cost time and money, although good online meeting tools bring these costs way down. One-on-one elicitation enables the analysts to better plumb experts' reasoning. Supervised plenary elicitation in which experts meet, discuss, and then individually perform the elicitation offer advantages of speed and disadvantages in loss of individual engagement.

Sending a questionnaire in the mail to a large set of experts in the hope that a fair number will respond is discouraged for purposes of uncertainty quantification. Expert surveys should be sharply distinguished from structured expert judgment.

Despite all this, the most difficult hurdle is the second: finding qualified analysts. Mathematicians, statisticians, engineers and scientists know that the Classical Model is not a heavy lift[5]. Many have conducted successful studies in their chosen fields. The analyst must be able to explain the method to the experts and to the problem owners, so that they in turn can explain it up the chain. If a problem owner is unable to explain the method to his/her superiors, (s)he is unlikely to adopt it. The analyst must be comfortable with certain relevant concepts such as statistical likelihood, p values, Shannon information, scoring rules, distributions, densities, quantiles, etc. Some knowledge of foundations is needed to explain why uncertainty is represented as subjective probability and not as fuzziness, imprecision, degree of possibility, certainty factors, to name a few[6]. Writing up the results in a clear an accurate fashion requires more than a nodding acquaintance with all these concepts.

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[5] Cooke (2015) esp. the supplementary online information is written to bring neophytes up to speed. The TU Delft launched a free online course on expert judgment in October 2019.

[6] See (Cooke 2015) for a foundational discussion targeting potential analysts.

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Appendix 1. Classical Model Performance Measures and Combination

based on courses given for NASA, FAA and various companies and labs from Cooke, R.M. *Experts in Uncertainty*, Oxford University Press, 1991

Online sources:

website for Structured Expert Judgment http://www.cooke-aspinall.net/

Wiki page: https://en.wikipedia.org/wiki/Structured_expert_judgment:_the_classical_model

Selected publications

- **REEP** Validation for Classical Model
- **RESS** Cross Validation extensive SOM on validation, aggregation
- **Comp & OR** Quantifying Info Security Risks
- **PNAS** SEJ ice sheets SOM SEJ data and elicitation protocol
- **PLoS 1** Evaluation of Performance: WHO SEJ study of global burden of disease
- **Elementa** Stormwater Management in Chesapeake Bay; extensive SOM elicitation protocol
- **Cons Bio** Impacts of Asian Carp Invasion Lake Erie - SEJ SOM
- **RFF** SEJ Breast Feeding and IQ & AStA
- **RFF** SEJ IQ and Earnings
- **Nature Climate Change:** Messaging climate Uncertainty with extensive **SOM** on representation of uncertainty

Videos

- **Intro to SEJ (10 mins)**
- **The Confidence Trap (10 mins)**
- **Ice sheets (11 mins)**
- **Validation (25 mins)**

Online Course

SEJ TU Delft

There are two generic, quantitative measures of performance, *statistical accuracy* (aka *calibration)* and *information*. Loosely, statistical accuracy denotes the statistical likelihood that a set of experimental results correspond, in a statistical sense, with an expert's assessments. Information measures the degree to which a distribution is concentrated. To simplify the exposition we assume that the 5%, 50% and 95% values were elicited.

Statistical accuracy

Assume that each expert assess 5, 50 and 95 percentiles for each quantity. Each expert divides the range of possible values into 4 inter quantile intervals for which his/her probabilities are known, namely $p_1 =$

0.05: less than or equal to the 5% value, $p_2 = 0.45$: greater than the 5% value and less than or equal to the 50% value, etc.

If N quantities are assessed, each expert may be regarded as a statistical hypothesis, namely that each realization falls in one of the four inter-quantile intervals with probability vector

$$p = (0.05, 0.45, 0.45, 0.05).$$

Suppose we have realizations $x_1,...x_N$ of these quantities. We may then form the sample distribution of the expert's inter quantile intervals as:

$$s_1(e) = \#\{i \mid x_i \leq 5\%\ quantile\}/N$$
$$s_2(e) = \#\{i \mid 5\%\ quantile < x_i \leq 50\%\ quantile\}/N$$
$$s_3(e) = \#\{i \mid 50\%\ quantile < x_i \leq 95\%\ quantile\}/N$$
$$s_4(e) = \#\{i \mid 95\%\ quantile < x_i\}/N$$
$$s(e) = (s_1(e),...s_4(e))$$

Note that the sample distribution depends on the expert e. If the realizations are indeed drawn independently from a distribution with quantiles as stated by the expert then the quantity

$$2NI(s(e)|\, p) = 2N \sum_{i=1..4} s_i \ln(s_i / p_i) \qquad \text{(A1.1)}$$

is asymptotically distributed as a chi-square variable with 3 degrees of freedom. This is the so-called likelihood ratio statistic, and $I(s \mid p)$ is the relative information of distribution s with respect to p. If we extract the leading term of the logarithm we obtain the familiar chi-square test statistic for goodness of fit. There are advantages in using the form in (A1.1) (Cooke 1991).

If after a few realizations the expert were to see that all realization fell outside his 90% central confidence intervals, he might conclude that these intervals were too narrow and might broaden them on subsequent assessments. This means that for this expert the uncertainty distributions are *not* independent, and he learns from the realizations. Expert learning is not a goal of an expert judgment study and his joint distribution is not elicited. Rather, the decision maker wants experts who do not need to learn from the elicitation. Hence the decision maker scores expert e as the statistical likelihood of the hypothesis

H_e: *"the inter quantile interval containing the true value for each variable is drawn independently from probability vector p."*

A simple test for this hypothesis uses the test statistic *(A1.1)*, and the likelihood, or p-value, or **statistical accuracy score** of this hypothesis, is:

$$Sa(e) = p\text{-}value = Prob\{2NI(s(e)|\, p) \geq r \mid H_e\}$$

where r is the value of *(A1.1)* based on the observed values $x_1,...x_N$. It is the probability under hypothesis H_e that a deviation at least as great as r should be observed on N realizations if H_e were true. Statistical accuracy scores are absolute and can be compared across studies. However, before doing so, it is appropriate to equalize the power of the different hypothesis tests by equalizing the effective number of realizations. To compare scores on two data sets with N and N' *realizations*, we simply use the minimum of N and N' in *(A1.1)*, without changing the sample distribution s. In some cases involving multiple realizations of one and the same assessment, the effective number of seed variables is based on the number of assessments and not the number of realizations.

Although the statistical accuracy score uses the language of simple hypothesis testing, it must be emphasized that we are not rejecting expert-hypotheses; rather we are using this language to measure the degree to which the data supports the hypothesis that the expert's probabilities are accurate. Low scores, near zero, mean that it is unlikely that the expert's probabilities are correct.

Information

The second scoring variable is information. Loosely, the information in a distribution is the degree to which the distribution is concentrated. Information cannot be measured absolutely, but only with respect to a background measure. Being concentrated or "spread out" is measured relative to some other distribution.

Measuring information requires associating a density to each variable based on each expert's quantile assessments. To do this, we use the unique density that complies with the experts' quantiles and is minimally informative with respect to a background measure. This density can easily be found with the method of Lagrange multipliers. For a uniform background measure, the density is constant between the assessed quantiles, and is such that the total mass between the quantiles agrees with p. The background measure is not elicited from experts as indeed it must be the same for all experts; instead it is chosen by the analyst.

The uniform and log-uniform background measures require an *intrinsic range* on which these measures are concentrated. The classical model implements the so-called k% overshoot rule: for each item we consider the smallest interval $I = [L, U]$ containing all the assessed quantiles of all experts and the realization, if known. This interval is extended to

$$I^* = [L^*, U^*]; L^* = L - k(U-L)/100; \quad U^* = U + k(U-L)/100.$$

The value of k is chosen by the analyst. A large value of k tends to make all experts look quite informative, and tends to suppress the relative differences in information scores. The **information score** of expert e on assessments for uncertain quantities 1…N is

$$Inf(e) = \text{Average Relative information wrt Background} = (1/N) \ \Sigma_{i=1..N} I(f_{e,i} \mid g_i)$$

where g_i is the background density for variable i and $f_{e,i}$ is expert e's density for item i. This is proportional to the relative information of the expert's joint distribution given the background, under the assumption that the variables are independent. As with statistical accuracy, the assumption of independence here reflects a desideratum of the decision maker and not an elicited feature of the expert's joint distribution. The information score does not depend on the realizations. An expert can give himself a high information score by choosing his quantiles very close together.

Evidently, the information score of e depends on the intrinsic range and on the assessments of the other experts. Hence, information scores cannot be compared across studies.

Of course, other measures of concentrated-ness could be contemplated. The above information score is chosen because it is

- familiar
- tail insensitive
- scale invariant
- slow

The latter property means that relative information is a slow function; large changes in the expert assessments produce only modest changes in the information score. This contrasts with the likelihood function in the statistical accuracy score, which is a very fast function. This causes the product of statistical accuracy and information to be driven by the statistical accuracy score. It also means that modest changes in informativeness correspond to sizeable changes in the distributions. Increasing informativeness by a factor *2* roughly corresponds to halving the distance between the 95 and 5 percentiles.

Combination: Decision Maker

The **combined score** of expert e will serve as an (unnormalized) weight for e:

$$w_\alpha(e) = Sa(e) \times Inf(e) \times 1_\alpha(Sa(e) \geq \alpha), \qquad (A1.2)$$

where $1_\alpha(Sa(e)) = 1$ if $Sa(e) \geq \alpha$, and is zero otherwise. The combined score thus depends on α. If $Sa(e)$ falls below cut-off level α expert e is unweighted. The presence of a cut-off level is imposed by the requirement that the combined score be an asymptotically strictly proper scoring rule. That is, an expert maximizes his/her long run expected score by and only by ensuring that his probabilities $p = (0.05, 0.45, 0.45, 0.05)$ correspond to his/her true beliefs. α is similar to a significance level in simple hypothesis testing, but its origin is indeed different. The goal of scoring is not to "reject" hypotheses, but to measure "goodness" with a strictly proper scoring rule.

A combination of expert assessments is called a "decision maker" (DM). All decision makers discussed here are examples of linear pooling. The classical model is essentially a method for deriving weights in a linear pool. "Good expertise" corresponds to good statistical accuracy (high statistical likelihood, high p-value) and high information. We want weights which reward good expertise and which pass these virtues on to the decision maker.

The reward aspect of weights is very important. We could simply solve the following optimization problem: find a set of weights such that the linear pool under these weights maximizes the product of statistical accuracy and information. Solving this problem on real data, one finds that the weights do not generally reflect the performance of the individual experts. As we do not want an expert's influence on the decision maker to appear haphazard, and we do not want to encourage experts to game the system by tilting their assessments to achieve a desired outcome, we must impose a strictly scoring rule constraint on the weighing scheme.

The scoring rule constraint requires the term $1_\alpha(\text{statistical accuracy score})$, but does not say what value of α we should choose. Therefore, we choose α so as to maximize the combined score of the resulting decision maker. Let $DM_\alpha(i)$ be the result of linear pooling for item i with weights proportional to (*A1*.2):

$$DM_\alpha(i) = \Sigma_{e=1,\ldots E}\, w_\alpha(e)\, f_{e,i} \;/\; \Sigma_{e=1,\ldots E}\, w_\alpha(e) \qquad (A1.3)$$

The *optimized global weight DM* is DM_{α^*} where α^* maximizes

$$\text{statistical accuracy score}(DM_\alpha) \times \text{information score}(DM_\alpha). \qquad (A1.4)$$

This weight is termed global because the information score is based on all the assessed seed items

A variation on this scheme allows a different set of weights to be used for each item. This is accomplished by using information scores for each item rather than the average information score:

$$w_\alpha(e,i) = 1_\alpha(\text{statistical accuracy score}) \times \text{statistical accuracy score}(e) \times I(f_{e,i} \mid g_i) \qquad (A1.5)$$

For each α we define the Item weight IDM_α for item i as

$$IDM_\alpha(i) = \Sigma_{e=1,\ldots E}\, w_\alpha(e,i)\, f_{e,i} \;/\; \Sigma_{e=1,\ldots E}\, w_\alpha(e,i) \qquad (A1.6)$$

The *optimized item weight DM* is IDM_{α^*} where α^* maximizes

$$\text{statistical accuracy score}(IDM_\alpha) \times \text{information score}(IDM_\alpha). \qquad (A1.7)$$

The non-optimized versions of the global and item weight DM's are obtained simply by setting $\alpha = 0$.

Item weights are potentially more attractive as they allow an expert to up- or down- weight him/herself for individual items according to how much (s)he feels (s)he knows about that item. "knowing less" means choosing quantiles further apart and lowering the information score for that item. Of course, good performance of item weights requires that experts can perform this up- down weighting successfully. Anecdotal evidence suggests that item weights improve over global weights as the experts receive more training in probabilistic assessment. Both item and global weights can be pithily described as optimal weights under a strictly proper scoring rule constraint. In both global and item weights statistical accuracy dominates over information, information serves to modulate between more or less equally well calibrated experts. Definitions and proofs of these scoring rule properties are found in Cooke, R.M. *Experts in Uncertainty*, Oxford University Press, 1991.

Since any combination of expert distributions yields assessments for the seed variables, any combination can be evaluated on the seed variables. In particular, we can compute the statistical accuracy and the information of any proposed decision maker. We should hope that the decision maker would perform better than the result of simple averaging of distributions, called the *equal weight DM*, and we should also hope that the proposed DM is not worse than the best expert in the panel. The global and item weight DM's discussed above (optimized or not) are *Performance based DM's.* In general the optimized global weight DM is used, unless the optimized item weight DM is markedly superior.

The optimization in *(A1.5)* and *(A1.7)* often causes experts to be unweighted, even experts with good scores. Such experts are not "rejected;" unweighting simply means that their input is already captured by a smaller subset of experts. Their value to the whole study is brought out in studying the robustness of the optimal *DM* under loss of experts.

APPENDIX 2. In Sample Validation

The histogram of calibration variables and the graph of *p*-value scores (statistical accuracy scores) are given in Figure A2.1.

Figure A2.1*: Calibration frequencies (left) for all 526 post-2006 experts who assessed all calibration variables and p-value scores (right) for all 530 experts not accounting for different numbers of assessed calibration variables. This includes 4 experts who skipped some calibration variables. The traditional 5% threshold for simple hypothesis testing is given as a red line.*

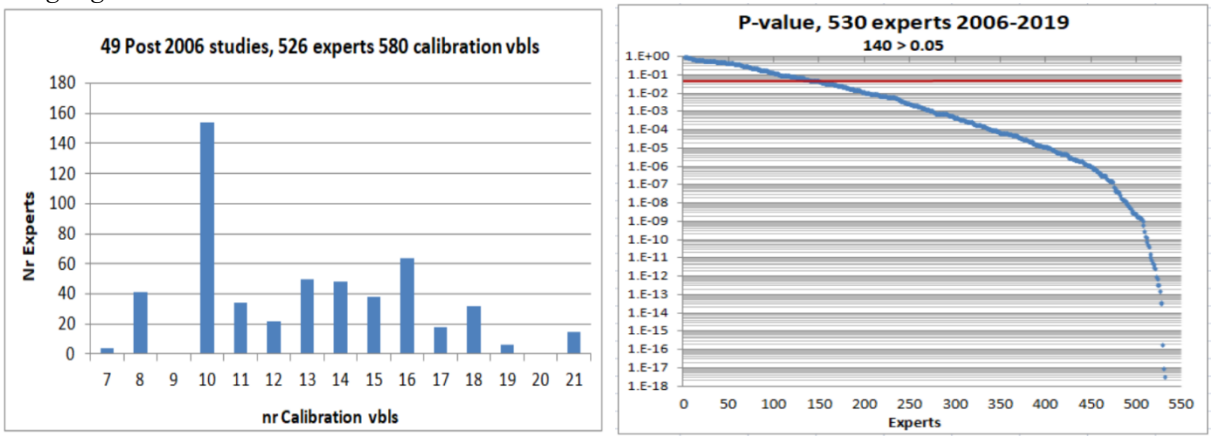
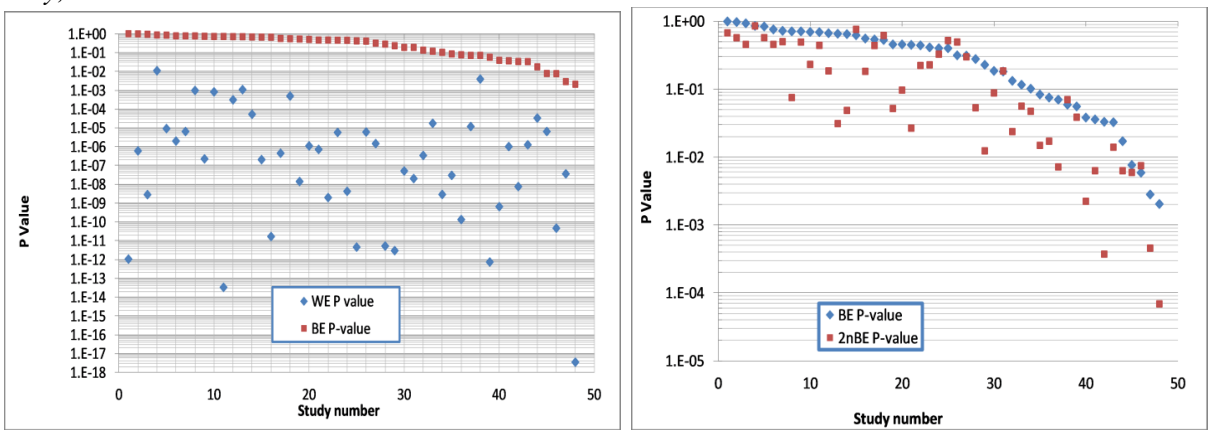


Figure A2.2 shows the best and worst *p-value*s per study for all *49* post-2006 studies (left) and the two best performing experts (right). There are generally *4* or more orders of magnitude in statistical accuracy scores between the best and worst expert per study. Despite the fact that only *140* of the *530* experts would not be rejected as statistical hypothesis at the *5%* level on simple hypothesis tests, most studies have one or even two statistically acceptable experts.

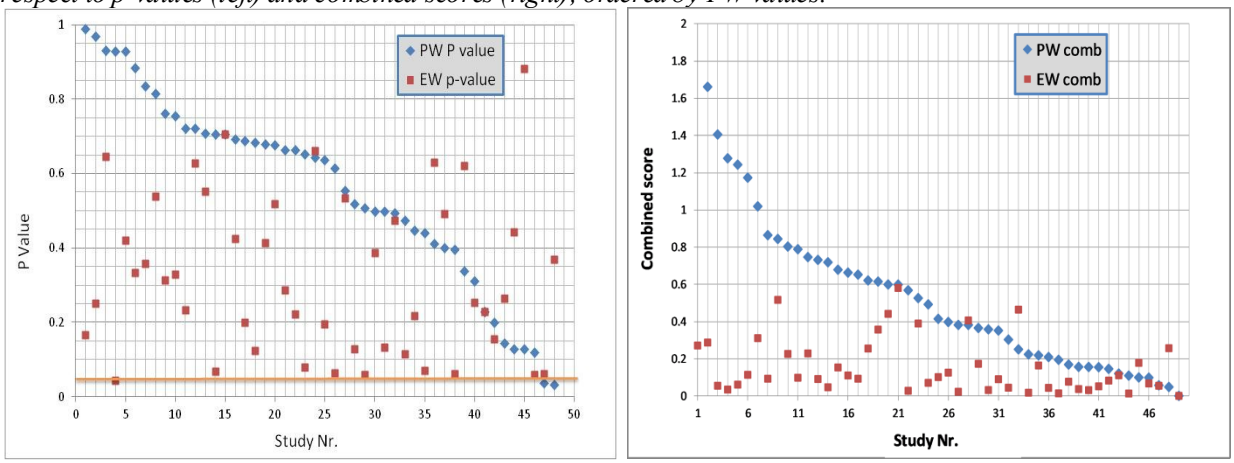
Figure A2.2: Best and worst expert (BE and WE, respectively) p-values or Statistical Accuracy (left) and p-values of best two experts in terms unnormalized weight (combined score of statistical accuracy × informativeness) (right), per study, 2006-2019 data.



Note. Studies are ordered with respect to best experts' statistical accuracy scores in both plots.

Comparing *PW* and *EW* decision makers on the data used to initialize the performance weighting is "in-sample validation". Figure A2.3 shows the in-sample results for the 2006-2019 data. The combined score is the product of the (dimensionless) statistical accuracy and informativeness scores. The left graph shows that *PW* and *EW* have roughly the same number of studies below the conventional 5% rejection threshold, though *PW* tends to be higher. The right graph adds the informativeness factor and boosts the in-sample superiority of *PW* over *EW*.

Figure A2.3: In-sample comparison of performance weighted (PW) and equal weighted (EW) decision makers with respect to p-values (left) and combined scores (right), ordered by PW values.



Appendix 3 Data and Data References

Post-2006 Expert Data

In addition to the *33* post-2006 studies studied in (Colson and Cooke 2017; 2018), 16 post 2016 studies have been added, as summarized in Table A3.1.

Table A3. 1 Expert judgment studies are shown with the number of calibration variables and experts, post-2006 (post-2016 bolded), asterisks denote studies where variables were excluded in section 5 owing to equality of prediction and realization. Four experts who did not assess all calibration variables are excluded.

Study	# Expert	# Calib Vbls	Subject
Arkansas	4	10	Grant effectiveness, child health insurance enrollment
arsenic	9	10	Air quality levels for arsenic
ATCEP	5	10	Air traffic controllers human Error
BFIQ	7	11	**Breastfeeding and IQ**
biol_agents*	12	12	Human dose-response curves for bioterror agents
Brexit_Food	10	10	**Food price change after Brexit**
CDC_all	48	14	**Global burden of disease**
CDC_ROI	20	10	Return on investment for CDC warnings
CoveringKids	5	10	Grant effectiveness, child health insurance enrollment
CREATE	7	10	Terrorism
CWD	14	10	Transmission risks: chronic wasting disease from deer to humans
Daniela	4	7	Fire prevention and control
dcpn_Fistula	8	10	Effectiveness of obstetric fistula repair
eBBP**	14	15	XMRV blood/tissue infection transmission risks
effErup	14	8	Icelandic fissure eruptions: source characterization
Erie	10	15	Establishment of Asian Carp in Lake Erie
FCEP	5	8	Flight crew human rror
Florida	7	10	Grant effectiveness, child health insurance enrollment
France	5	10	**Future antimicrobial eesistance in France**
Geopolit	9	16	**Geopolitics**
Gerstenberger	12	13	Probabilistic Seismic-Hazard Model for Canterbury
GL_NIS	9	13	Costs of invasive species in Great Lakes
Goodheart	6	10	Airport safety
Hemophilia	18	8	Hemophilia
Ice_2012	10	11	Sea level rise from Ice Sheets melting due to global warming
ICE_2018	20	16	**Future see level rise**
Illinois	5	10	Grant effectiveness, child health insurance enrollment
IQEarnings	8	11	**Effect IQ in India on present value of lifetime earnings**
Italy	4	8	**Future antimicrobial eesistance in Italy**
Liander	11	10	Underground cast iron gas-lines
Nebraska	4	10	Grant effectiveness, child health insurance enrollment
Obesity	4	10	Grant effectiveness, childhood obesity
PHAC	10	12	Additional CWD factors
political_violence	15	21	**Political violence**
puig_gdp	9	13	**Emission forecasts from Mexico**
puig_oil	6	19	**Oil emissions and prices**
sanDiego	7	10	Effectiveness of surgical procedures
Sheep	14	15	Risk management policy for sheep scab control
spain	5	10	Future antimicrobial esistance in Spain
SPEED	14	16	**Volcano hazards (Vesuvius & Campi Flegrei, Italy)**
Tadini_Clermont*	12	13	Somma-Vesuvio volcanic complex geodatabase
Tadini_Quito*	8	13	**Volcanic risk**
TdC*	18	17	**Volcano hazards (Tristan da Cunha)**
Tobacco*	7	10	Grant effectiveness, childhood obesity
TOPAZ*	21	16	Tectonic hazards for radioactive wastes siting in Japan
UK	6	10	**Future antimicrobial esistance in UK**
UMD	9	11	Nitrogen removal in Chesapeake Bay
USGS	32	18	**Volcanos**
Washington	5	10	Grant effectiveness, child health insurance enrollment
* contains one variable where *PW*iMD=*true value*			
** contains 4 variables where *PW*iMD=*true value*			

Note. # Calib Vbls refers to the number of calibration variables that each expert assessed in a study. Studies are sorted in alphabetical order. Variables where *PW*iMD=*rls* are excluded in section 5, since they cannot be rendered scale invariant.

Table A3.3 References for post 2006 expert studies.

Study Name	Reference
Arkansas	Attribution study for Robert Wood Johnson Covering Kids & Families in Pennsylvania, Washington, Nebraska, Illinois, Arkansas, and Florida, conducted by Center for Disease Dynamics, Economics & Policy, 2012.
arsenic	Hanzich, J.M. (2007) Achieving Consensus: An Analysis Of Methods To Synthesize Epidemiological Data For Use In Law And Policy. Department of Public Health & Primary Care, Institute Of Public Health, University of Cambridge; unpublished MPhil thesis, 66pp + appendices.
ATCEP	Morales-Nápoles, O., Kurowicka, D., & Cooke, R. (2008). EEMCS final report for the causal modeling for air transport *S*afety (CATS) project.
BFIQ	Colson, A. Cooke, R.M., Lutter, Randall, (2016) How Does Breastfeeding Affect IQ? Applying the Classical Model of Structured Expert Judgment, Resources for the Future, RFF DP16-28 http://www.rff.org/research/publications/how-does-breastfeeding-affect-iq-applying-classical-model-structured-expert
Biol Agents	Aspinall & Associates (2006). REBA Elicitation. Commercial-in-confidence report, 26pp.
Brexit_food	Barons MJ, Aspinall W. (2020) Anticipated impacts of Brexit scenarios on UK food prices and implications for policies on poverty and health: a structured expert judgement approach. BMJ Open 2020;10:e032376. doi:10.1136/ bmjopen-2019-032376
CDC ALL	past clearance, publication in preparation
cdc-roi	Colson, Abigail R., M.A. Cohen, S. Regmi, A. Nandi, R. Laxminarayan (2015) "Structured Expert Judgment for Informing the Return on Investment in Surveillance: The Case of Environmental Public Health Tracking." Working Paper. Center for Disease Dynamics, Economics & Policy.
CoveringKids	Attribution study for Robert Wood Johnson Covering Kids & Families in Pennsylvania, Washington, Nebraska, Illinois, Arkansas, and Florida, conducted by Center for Disease Dynamics, Economics & Policy, 2012.
create	Bier V.M, Kosanoglu, F, Shin J, unpublished data, nd.
CWD	Tyshenko, M.G., Elsaadany, S., Oraby, T., Darshan, S., Catford, A., Aspinall, W., Cooke, R. and Krewski, D. (2012) Expert judgement and re-elicitation for prion disease risk uncertainties. *International Journal of Risk Assessment and Management*, 16(1-3), 48-77. doi:10.1504/IJRAM.2012.047552
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